Contrast Consistent Search for Large Vision Language Models

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Abstract

Large Vision–Language Models (LVLMs) often struggle to produce accurate and reliable image descriptions, frequently hallucinating or omitting key details related to object detection, attribute recognition, and spatial relationships. Traditional methods to reduce such errors rely on extensive datasets, complex architectural redesigns, or human feedback, yet these approaches often yield inconsistent performance. In this work, we introduce a novel probe-optimization framework that extends the Contrast Consistent Search (CCS) method to LVLMs, enabling unsupervised enhancement of image understanding. Our CCS-based approach extracts latent knowledge from internal model activations and enforces compositional consistency constraints to promote truthful and reliable outputs, even without ground-truth supervision. Experimental results show that Qwen2.5 achieves the highest overall accuracy (83.3%), followed closely by Qwen2 (83.2%), while LLaVA trails at 77.2%. Across all dimensions, the unsupervised models outperform their supervised counterparts, demonstrating that internal representational consistency—rather than the scale of data or supervision—drives better reasoning. By allowing models to assess and refine their own internal representations, the proposed CCS framework narrows the gap between unsupervised and supervised LVLMs, improving trustworthiness in image description tasks.

Keywords

Contrast Consistent Search, Hallucinations, Image Description(s), Large Vision Language Models, Unsupervised Models, Supervised Models

1 Introduction

Large Language Models have exploded in the past years and have made exceptional progress when it comes to Natural Language Understanding (NLU) and Natural Language Generation (NLG) [13]. Building upon this foundation Multimodal Large Language Models (MLLMs) have recently emerged as a transformative paradigm in artificial intelligence, capable of integrating and reasoning across diverse data modalities such as text, images, and video [14] [17]. By extending the architecture and training philosophy of traditional Large Language Models (LLMs), MLLMs retain robust linguistic understanding while incorporating visual perception and grounding. These models have demonstrated impressive versatility across tasks such as image captioning, visual question answering, and text-to-image generation, thereby bridging the gap between vision and language in a unified framework [14].

Despite their remarkable progress, a persistent and critical issue that limits the reliability of MLLMs is hallucination. Hallucination occurs when a model describes objects, attributes, or relationships that are not present in the input image, or omits salient visual details altogether [13]. They occur when there is either too much/too little data. For instance, in image description tasks, models may inaccurately report an object's presence, misrepresent its colour, or distort spatial relationships among visual entities. These inconsistencies undermine the factual alignment between visual content and linguistic output [11], raising serious concerns for real-world applications that demand high trustworthiness.

Hence, training MLLMs to produce accurate and contextually grounded descriptions requires massive amounts of annotated data, which is often expensive, domain-specific, and difficult to scale. Even with such data, existing models struggle to consistently mitigate hallucinations, revealing inherent limitations in their ability to self-correct or reason about uncertainty. While some models also intend to use human feedback [10] to identify and penalize hallucinated outputs, this manual evaluation is costly, time-consuming, and can be potentially biased [14]. Moreover, human involvement introduces indirect incentives that may distort a model's internal learning dynamics. Thus, there is a compelling need for label-free approaches that enable MLLMs to evaluate the reliability of their own predictions.

LVLMs a subclass of MLLMs that utilises textual and visual inputs for generation and understanding suffers from similar impediments showcasing hallucinations and inconsistencies [16]. To address this problem, we extend the CCS framework—originally designed for text-only LLMs [7] to LVLMs. CCS is an unsupervised method to extract knowledge internally using the hidden activations. These intermediate layers hold the truth values that represent what the model inherently believes in, rather than what is outputs. CCS offers a valuable mechanism for improving LVLMs because it identifies a linear direction in the model's activation space that distinguishes true from false statements based on internal consistency, without relying on external supervision or labelled data. By maximising contrast and internal coherence, CCS leverages the model's own latent knowledge rather than its final outputs. This makes it especially useful for LVLMs, where hallucinations are common and supervision is costly or limited. In essence, CCS provides a way to guide LVLMs toward factual correctness across diverse topics by aligning their internal reasoning, making it a powerful tool for reducing hallucinations and enhancing reliability without extensive retraining or major architectural changes [7].

In this work, following contributions are made:

- Self-assessment instead of supervision is introduced: A new perspective is proposed in which vision–language misalignment is addressed by allowing LVLMs to self-assess their visual understanding through internal consistency, rather than being guided by external labels or human-defined truths.
- Internal knowledge is leveraged to determine truthfulness: The model's implicit beliefs, learned during multimodal training, are utilised under the assumption that truthful internal representations should exhibit logical and contrastive consistency—properties unlikely to arise from hallucinations or random associations.
- Reliability in description is reinforced: When applied to image captioning tasks, internal reliability scores are generated by the model, allowing more consistent and hallucination-free visual descriptions to be produced, even in the absence of labelled data.
- Progress toward self-aware and trustworthy LVLMs is achieved: By enabling internal reasoning about the correctness of outputs, trustworthiness is enhanced, gaps between shallow and deep features can be visualised, and a step is taken toward self-aware multimodal intelligence.

2 Related Work

2.1 Understanding the inner working of an LLM

Scientists have been getting really interested in figuring out why large language models sometimes make stuff up, and they are coming at it from different angles. At first, researchers would just check if the AI's answers were consistent or try to measure how confident it seemed, but this didn't really get to the heart of why these models create fake information in the first place. Then some scientists came up with something called Contrast-Consistent Search(CCS), which was unique and interesting because instead of just looking at what the model says, they investigated what's actually happening inside the model's brain to spot when it might lie. More recently, in 2025, some researchers introduced something called Steer LLM Latents [15] for Hallucination Detection. They figured out how to not just catch these lies but actually encourage the model's brain to better separate true stuff from made-up stuff using what they call a steering vector. What's exciting about all this work is that instead of just trying to catch problems after they happen, focus is towards starting to think about how the AI's brain actually differentiates between truth and false notions.

2.2 Cross-Modal Consistency and Representation Alignment

Another paper that looks at how different types of information stay consistent with each other is called "MedCLIP: Contrastive Learning from Unpaired Medical Images and Text" [20]. Most systems that work with both images and text need matching text descriptions. But MedCLIP is unique in the sense it works with medical images and clinical reports that aren't matched up perfectly. The system figures out how to connect visual information with text information using a learning approach that brings together things that make sense when they are together. This means it doesn't need huge databases of perfectly labeled examples that are really hard to get in medical fields, and it gets better at understanding connections between images and text. Even though this work focuses specifically on healthcare, the idea of making sure visual and text information works well together fits right in with what other systems like CCS are trying to do, they are all working toward keeping meaning consistent across different types of data.

2.3 Benchmarking and Controlled Evaluation

Early studies usually evaluated hallucinations using fixed image datasets . Tools like POPE and related assessments quantified mismatches between textual descriptions and images but they captured only major semantic errors. To improve this understanding, a new paper in 2024 [19] introduced VHTest, a framework that generates diverse hallucination instances that cover eight visual modes namely existence, color,counting, and position. By synthesizing adversarial examples with MLLM models revealed how MLLMs hallucinate when put under varied and controlled conditions, and showed that fine tuning such data reduces hallucination rates without harming general performance.

The structural solution proposed in [9] goes beyond benchmark-based methods. Through the use of a cross-graph Siamese network, their graph based concept puts together text and image as scene graphs which can be seen as a way to correlate the both of them. Mismatches in image-to-text or vice versa are detected by the approach by comparing objects, properties, and relations across the visual graphs. With this approach, hallucination detection is moved from pixel-level to interpretable, semantic, and easily comprehensible reasoning.

2.4 Domain-Specific and Reasoning-Focused Solutions

While multimodal hallucination is often studied in open-domain imagery, the problem also arises in text-only contexts. In the e-commerce domain, [12] addressed issues in product listings enriched by LLMs. Their two-phase pipeline first uses lexical and semantic screening to flag potential hallucinations and then employs an LLM for selective validation. This design balances both scalability and precision, showing how hallucination detection can be used at industrial scale without direct access to model internals. These are beneficial for a lot of e-commerce companies which do not have the necessary means to have access to model internals.

Other works explore related multimodal reasoning challenges. A paper on LDRE [18] proposed the LDRE method for zero-shot included image retrieval, using LLMs for actions such as divergent reasoning. The system generates many candidate captions illustrating different meanings of a reference image and text modification and then joins them through a concept called semantic weighting. This approach highlights the reasoning diversity in aligning language and vision.

All of these studies point towards a transition from surface-level error counting toward deeper, model-aware understanding. Recent work combines benchmark generation (VHTest) [19] , adversarial probing (GHOST) [6] , structured graph to build a more complete picture of multimodal hallucination.

3 Problem Statement

General image descriptions generated from LVLMs suffer from inconsistencies due to hallucinations mainly across dimensions object detection, attribute recognition and spatial relationships. We try to address the root of the problem by identifying the weak dimension(s) in the respective model(s) and simultaneously giving insights on the model dimensions that require improvement (enabling LVLMs to train better and faster). However, instead of relying on methods that change their output based on a reward mechanism, an effort was made to extract knowledge that is present in the hidden states of the model.

This is done by using a pretrained LVLM and a set of questions from $q_1, q_2, \ldots, q_n$ for dimensions d_1, d_2, d_3 representing object detection, attribute recognition and spatial relationships, respectively. The goal here is to answer the questions q_i in a well-defined way, i.e., "Yes" or "No" (as shown in Figure 1) without using explicit external supervision.

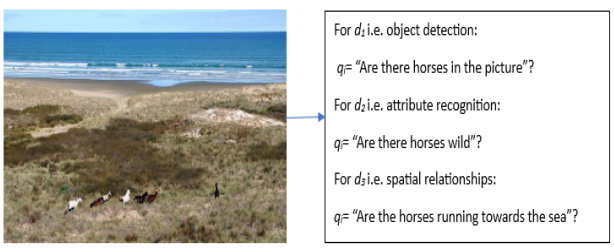

Figure 1: An example of dimension-specific questions for an image.

4 Methodology

4.1 Method: CCS (adapted for images)

In order to move towards a solution, the original CCS principle [7], based on truth's consistency under negation, is adapted for images to find a direction in the activation space that keeps answers consistent across negated question pairs. It follows a four-step approach of working:

- Each question q_i is answered as both "Yes" (x_{pos}) and "No" (x_{neg}).
- Corresponding feature representations, $h(x_{pos})$ and $h(x_{neg})$, are computed.
- Probabilities are assigned to each answer, p_{pos} and p_{neg}.
- The probability mappings are optimised to enhance both confidence and consistency.

The overall structure of the CCS (as observed in Figure 2) has the following structure i.e. the input to CCS is set of binary questions from $q_1, q_2, \ldots, q_n$ for dimensions d_1, d_2, d_3 (representing object detection, attribute recognition and spatial relationship resp.) such that q_i only has access to feature representations $h(x)$. In short, CCS does not train or modify the LVLM. It only looks at the model's internal feature representations (hidden activations) and uses them to answer binary questions in a logically consistent way, without using any labels or changing the model's weights.

4.2 Adapting CCS in LVLMs

To adapt CCS to LVLMs, the internal reasoning of the model is used to evaluate whether its visual interpretation is logically aligned with its linguistic output, without relying on external labels or supervision. The adaptation process consists of the following key stages:

4.2.1 *Input and Binary Questioning.*

- An image is provided to the LVLM.
- Instead of generating open-ended captions directly, the model is asked a series of binary, contrastive questions (Yes/No format).
- Each question is paired with its negation (contrast pair), ensuring that answers must satisfy **negation consistency**—if the model says "Yes" for a statement, it should say "No" for its logical negation.

4.2.2 *Feature Extraction and Internal Probing.*

- For each question pair $(x_{\text{pos}}, x_{\text{neg}})$, the LVLM's hidden activations are extracted using its feature extractor $h(x)$.
- These activations are normalised to ensure comparable scale and to avoid clustering of positive vs. negative answers in unrelated regions of the activation space.
- A linear probe followed by a sigmoid function maps these activations to probabilities, representing how likely the model internally believes each statement to be true.

4.2.3 *Consistency-Based Training Objective (Unsupervised).* CCS trains the probe not with labeled answers, but by enforcing consistency between contrast pairs:

- **Consistency Loss:** Ensures that $p(x_{\text{pos}}) \approx 1 - p(x_{\text{neg}})$.
- **Confidence Loss:** Prevents trivial solutions where both probabilities are equal to 0.5 by encouraging confident predictions.
- The total CCS loss is the sum of both losses averaged over all question pairs.

4.2.4 *Reliability Score and Final Output.*

- After optimisation, the model computes a **reliability score** (here i.e. the probability of an answer being yes) for each question:

$$p(q_i) = \frac{p(x_{\text{pos}}) + \left(1 - p(x_{\text{neg}})\right)}{2}$$

- If $p(q_i) > 0.5$, the answer is considered "Yes"; otherwise, "No".
- This produces a multi-dimensional internal reliability vector, representing how confident and consistent the model is across different visual aspects such as object detection, attributes and spatial relationships.
- The results are finally compared against a ceiling model to evaluate accuracy.

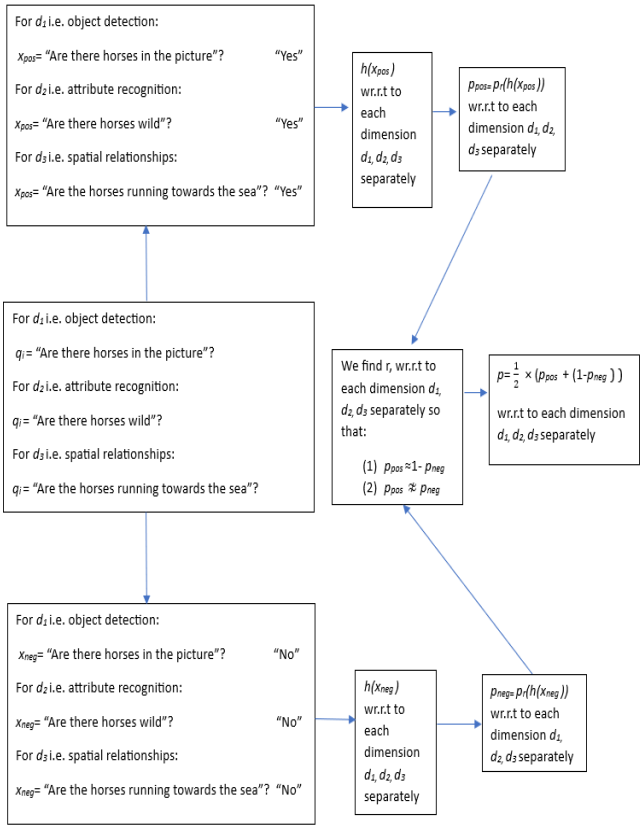

Figure 2: Given a set of questions q_i for dimensions $d_1, d_2, d3$, each question is answered with both a "Yes" and "No", followed by the extraction of internal model activations represented by $h(x_{pos})$ and $h(x_{neg})$ respectively. Further, each activation is mapped to the probability of it being true (as p_{pos} and p_{neg}), followed by optimisation of loss to make probabilities consistent and confident. In the end, for each question, the probability of an answer being "Yes" is estimated.

5 Experimental Setup

A comprehensive view of the experimental setup can be found at https://github.com/Kaolver/VisionCCS.git.

Models: One of the important decisions was to use the appropriate LVLM models to get the desired results. After going through the pros and cons of multiple models we arrived at using LLaVA [1] , Qwen2 [2] and Qwen2.5 [3] as the desired LVLM's. LLaVA is a type of AI model that can interpret images and comprehend language, then integrate both to respond to inquiries, explain visuals, or engage in intelligent discussions about what it observes. Qwen on the other hand is an open-source multimodal AI that combines image and text understanding for vision-language applications like captioning, visual question answering, document analysis, and object localization. Its chat version, Qwen-VL-Chat, supports multi-image, bilingual conversation with high-resolution, fine-grained recognition. More sophisticated versions like Qwen-VL-Plus, Qwen-VL-Max, and Qwen3-VL include the capability to video, spatial reasoning, and long-context processing, positioning Qwen-VL as a strong, flexible alternative to models like GPT-4V for vision-language applications. Similarly Qwen2.5 also has many variations like the Qwen2.5-VL which is a vision language model, Qwen2.5-Math which can solves math problems described in images.

Data: 2 datasets were shortlisted for the experiment. One being MME [4] and the other being VQAv2 [5] . Only one of them could have been used. MME even though was more extensive had a skewed distribution thus the curve was not as desired. VQAv2 on the other hand had better distribution making for a better dataset overall for the experiment. The experiment consisted of 3 parts object detection, spatial recognition and attribute recognition. There were exactly 7200 rows of data. Since CCS is suitable for binary questions, the data points comprising of binary questions were selected. These questions had to be further categorised. Llama 3.1(8 billion parameters) was used to categorize each data point into 3 parts object detection, spatial recognition and attribute recognition. Any other data point that did not fall into any of these 3 was categorized as "other". 1323 data points went for object detection, 1030 for spatial recognition, 3410 for attribute recognition and 1437 as "other". The same can be viewed in Table 1.

Methods: CCS is an effective unsupervised learning method but it must be compared against a supervised method like logistic regression. This experiment aims to prove that an unsupervised learning method can show similar results to a supervised learning model such as logistic regression. One of the reasons that influenced the decision of choosing logistic regression is the fact that in the original CCS paper [8] logistic regression was used as well.

Implementation Pipeline: The tokenizer is used to process the inputs into a condensed form for the AI model to learn. In multimodal systems using the CCS idea, tokenization creates a string of visual tokens – these are continuous vectors that a language model can use. A vision encoder then chops up the image into patches and makes an embedding for each. Then, a light connector makes a dense token sequence in the LLM's embedding space by either compressing or aligning the embeddings. This goes in with the usual BPE-tokenized text. So, the CCS tokenizer changes images into tokens the model can understand as context, allowing CCS to check if the output text is consistent and matches the visual information.

The pooling mechanism functions by compressing the high-dimensional information from a model's hidden states into a concise vector. After the model processes text and pictures, it generates hidden states. The pooling then aggregates all those states into one vector. In this research and the original CCS implementations, last token pooling has been applied. Consequently, this process produces a robust summary of the model's internal representation. CCS subsequently utilizes these pooled representations to evaluate the coherence and consistency of the model's generated output.

The project was executed within a Python framework, leveraging the computational resources of the Snellius supercomputer. In particular, the gpu-a100 partition was utilized, specifically employing NVIDIA A100-SXM4-40GB GPUs (40960 MiB, driver 580.95.05). To optimize resource allocation, we also made use of the gpu-mig

Total	Object detection	spatial recognition	attribute recognition	Other
7200	1323	1030	3410	1437

Table 1: Data Distribution

partition, which leverages Multi-Instance GPU (MIG) technology to allow for fractional GPU scheduling, ranging from one-half to a full GPU instance.

On the software side, all models were implemented using the HuggingFace transformers library, specifically to load `llava-1.5-7b-hf`, `Qwen2-VL-7B-Instruct`, and `Qwen2.5-VL-7B-Instruct`.

6 Results

The results in Table 2 and Table 3 offer a clear, quantitative lens through which the relative strengths of LLaVA, Qwen2, and Qwen2.5 can be evaluated in the context of three key visual-language understanding dimensions—object detection, attribute recognition, and spatial relationship reasoning. Across all dimensions, the newer Qwen2 and Qwen2.5 models consistently outperform LLaVA, with Qwen2.5 achieving the highest accuracies overall: 84.8% for object detection, 85.5% for attribute recognition, and 79.6% for spatial reasoning, yielding an average accuracy of 83.3%. This reflects a clear improvement in both attribute understanding and relational reasoning between objects. By comparison, LLaVA achieves 79.0%, 77.0%, and 75.5% respectively (average 77.2%), highlighting a gap of roughly 6–7 percentage points across dimensions. This dimension-wise performance (as observed in Figure 3, Figure 4 and Figure 5 resp.)increase implies that successive Qwen versions exhibit stronger internal alignment between visual and linguistic modalities, with Qwen2.5 demonstrating the best understanding of what it sees and how it describes it.

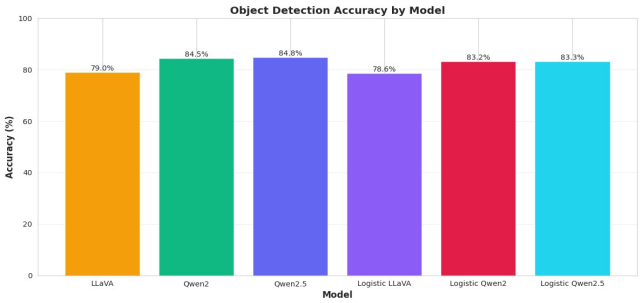

Figure 3: Object detection accuracy across models.

When considering average accuracy, the progression from 77.2% (LLaVA) to 83.2% (Qwen2) and finally 83.3% (Qwen2.5) captures a steady enhancement in holistic model performance (as observed in Figure 6). The average accuracy here shows how consistent the model is across different types of reasoning-seeing objects, describing their features, and understanding their relationships. The small difference between Qwen2 and Qwen2.5 means that Qwen2.5 became a bit more efficient and stable, but Qwen2 was already very strong in its internal understanding. This levelling off suggests that future improvements may not come from using more

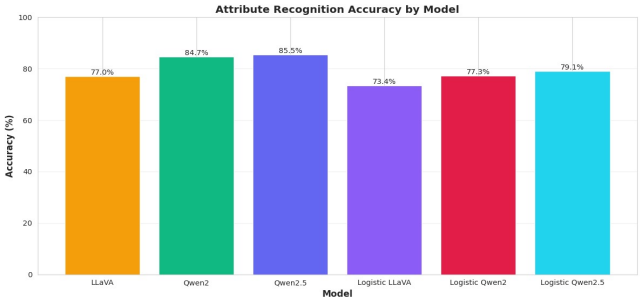

Figure 4: Attribute recognition accuracy across models.

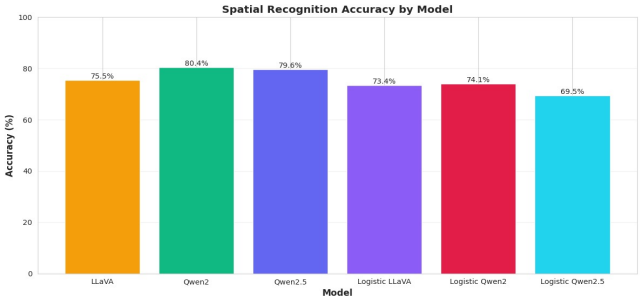

Figure 5: Spatial relationship accuracy across models.

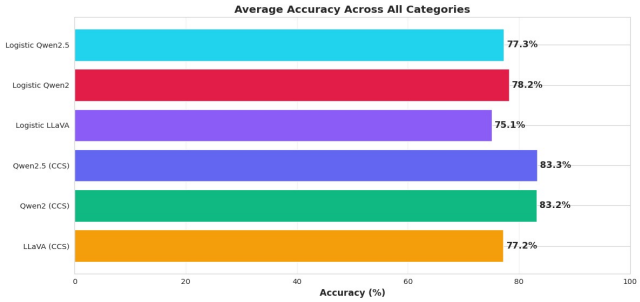

Figure 6: Overall comparison of accuracies across models.

data or supervision, but from making the model's internal reasoning more consistent, which is exactly what the CCS framework focuses on.

Examining the LR models, reveals an insightful pattern. For LLaVA, LR performance drops slightly across all tasks—78.6% for object detection, 73.4% for attribute recognition, and 73.4% for spatial relationships—resulting in a lower average (75.1%) than the CCS model. This decline indicates that introducing external supervision does not enhance the model's feature separability. For

Table 2: Comparing accuracy for object detection, attribute recognition and spatial relationship using LLaVA and Qwen w.r.t to the ceiling model, i.e. here Logistic Regression.

Model Name	Object Detection	Attribute Recognition	Spatial Relationships	Average Accuracy
LLaVA	79.0	77.0	75.5	77.2
Qwen2	84.5	84.7	80.4	83.2
Qwen2.5	84.8	85.5	79.6	83.3
Logistic Regression (based on LLaVA)	78.6	73.4	73.4	75.1
Logistic Regression (based on Qwen2)	83.2	77.3	74.1	78.2
Logistic Regression (based on Qwen2.5)	83.3	79.1	69.5	77.3

Table 3: Loss for object detection, attribute recognition and spatial relationship using LLaVA and Qwen, respectively.

Model Name	Object Detection	Attribute Recognition	Spatial Relationships
LLaVA	0.00087	0.00529	0.00135
Qwen2	0.00050	0.00326	0.00094
Qwen2.5	0.00044	0.00245	0.00092

Qwen2, LR achieves 83.2%, 77.3%, and 74.1% across dimensions (avg 78.2%), again trailing behind the unsupervised CCS model (83.2%). Similarly, Qwen2.5's supervised counterpart nearly matches the CCS model numerically (83.3%, 79.1%, 69.5%, avg 77.3%), but it does not perform as well. In all three cases, these results reveal that LVLMs' internal latent representations are already rich, discriminative, and self-organised, leaving little room for improvement through additional supervised mapping.

This distinction between the unsupervised and supervised forms of each model reinforces one of the core insights of the CCS approach—that meaningful representational alignment arises from within the model's learned feature geometry rather than through explicit label conditioning. The CCS method leverages this by enabling self-assessment: instead of rewarding models for producing externally validated outputs, CCS examines internal consistency across hidden states, thereby refining representations without introducing bias. The slight underperformance of logistic regression highlights how external supervision can unintentionally constrain the model's natural expressiveness, while CCS-style introspective training retains flexibility and promotes reliability.

The loss analysis (as observed in Table 3) further supports this conclusion. Lower loss values indicate stronger internal consistency and better alignment between visual and language representations. Here again, Qwen2 and Qwen2.5 outperform LLaVA across all tasks, with Qwen2.5 achieving the lowest overall losses (0.00044 for object detection, 0.00245 for attribute recognition, and 0.00092 for spatial reasoning). The reduction in loss magnitude from LLaVA to Qwen2.5 reflects improved contrastive consistency within latent spaces—an outcome that showcases that CCS explicitly optimises. In practical terms, lower loss values signal not only improved predictive stability but also reduced hallucination tendencies, as the model's internal representations become more logically consistent and self-reinforcing.

In summary, these results confirm that Qwen2.5 leads in accuracy and internal stability, followed by Qwen2 and then LLaVA, with all unsupervised LVLMs outperforming their supervised counterparts. The differences in performance between the models mainly come from how consistent their internal representations are, not from how much data or supervision they have. By using the model's own internal patterns to check and strengthen itself, CCS helps close the gap between unsupervised and supervised systems. This makes models like Qwen more reliable, easier to interpret, and stable even with limited training data. The combination of low loss and high accuracy across Qwen versions proves CCS's main point: improving how a model aligns its own internal knowledge is more important than adding more external supervision for better multimodal understanding.

7 Limitations and Future Work

During the process of doing this project, it turned out to be challenging to find a dataset containing the kind of questions this experiment required for the implementation of CCS. Initially, an LLM had to be deployed to categorise the questions into 3 subcategories: object detection, spatial recognition and attribute recognition. The dataset could not be as comprehensive, with around 7200 data points instead of the available 17,000, which may have reduced the accuracy of the results.

Another challenge was the selection of an appropriate dataset. Many datasets have poor distribution, thus giving undesirable results. The choice for this project was between the MME and VQA datasets. The VQA dataset was ultimately chosen because of better distribution.

Future improvements might include a larger dataset having a more diverse range of questions more questions for each of the categories, thus improving the overall accuracy.

Due to dataset limitations, we were unable to implement an additional unsupervised constraint that we considered: compositional consistency across dimensions. This approach would check for logical consistency between categories. For example, if we had the following spatial question: "the dog is to the left of the cat," both dog and cat must also be detected in the object recognition dimension. However, the dataset structure with typically one question per

image made this method not feasible to explore. Implementing compositional constraints would require datasets with comprehensive question coverage per image across all three dimensions. This cross-dimensional consistency check would align with CCS's philosophy of learning from logical structure rather than labels, potentially improving accuracy by adding another layer of unsupervised signal beyond the existing positive-negative statement consistency.

After observing the lower perfomance of Logistic Regression, an alternative method (Linear Probe) was considered and experimented with. Although it was not a part of the methodology of this research, we discovered that it performs better than Logistic Regression. Furthermore, Logistic Regression may be less suitable for multimodal vision-language representations compared to text-only models. This baseline was appropriate in the original CCS implementation, which focused only on text processing with language models. Hence, we believe that alternative supervised methods, such as the Linear Probe, should be explored to achieve a more precise comparison for vision-language tasks.

8 Conclusion

LVLMs are predicted to be used heavily in the years to come. This means that the accuracy of LVLMs is vital to us. This project was a detailed and comprehensive implementation of the CCS method, which extracts information from the internal workings of an LLM. This method, even though unsupervised, was found to outperform the supervised Logistic Regression, which was quite unexpected. Although the results were surprising, they had some merit since Logistic Regression may not suit multimodal applications of CCS as a comparison-based model. The fact that this experiment was able to exceed 80% accuracy forms a strong basis to prove that CCS can be very effective for this image-based application as well, not only for text-based processing.

AI Statement

The research project was largely a result of combined human effort; however, during this research project, ChatGPT was sometimes used to assist in transforming sentences to academic style. ChatGPT was also used to brainstorm some programming principles and practices for the realisation of the project, and to find and resolve some bugs in the code. All ideas, structures, and decisions in this report and code were extensively studied in advance. Any AI suggestion was carefully reviewed before incorporation into the project. We as a group, take full responsibility for the content of this work.

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